

IQRA TABASSUM

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Summary

ML researcher with a Scopus-indexed publication and production software engineering experience. Skilled in building end-to-end ML/DL pipelines, data preprocessing, and backend systems using Python, PyTorch, and TensorFlow, ready to bridge research and engineering in real-world AI products.

Core Competencies

ML/AI Frameworks: PyTorch, TensorFlow, Scikit-learn, LSTM, Autoencoders, Anomaly Detection, LLM

Languages: Python, JavaScript, SQL, Reactjs

Research Skills: Supervised /Unsupervised Learning, Time-Series Analysis, Feature Engineering, Algorithm Design

Data Engineering: ETL Pipelines, Data Preprocessing, Signal Processing, Data Mining, Batch Processing

Cloud & DevOps: Linux, CI/CD, Testing

Publication

Unsupervised Anomaly Detection for Psychiatric Inpatients Using Wearable Sensor Data: A Real-Time Monitoring Framework for Clinical Risk Management—[KICS \(2026, Scopus-indexed\)](#).

Experience

Research Assistant

[Lab Prof.](#)

Asan, South Korea

03/2024 - 02-2026

- Designed and built a real-time anomaly detection system using wearable sensor data for early clinical risk monitoring, employing LSTM and Autoencoder architectures.
- Collected, cleaned, and preprocessed raw hospital and sensor datasets; performed feature engineering and time-series preparation for ML/DL pipelines at scale.
- Built, trained, and evaluated multiple unsupervised and supervised ML/DL models; implemented ensemble methods and comparative evaluation pipelines using Python, TensorFlow, and PyTorch.
- Published in [KICS \(2026, Scopus-indexed\)](#); funded by Korea Health Technology R&D Project (KHIDI), Ministry of Health & Welfare, Republic of Korea

Software Engineer

[NetSol](#)

Lahore, Pakistan

07/2022 - 02-2024

- Contributed to production-grade **enterprise financial** platforms for international clients, including systems supporting Mercedes-Benz financial and leasing operations.
- Developed backend **business logic**, MySQL database structures, and production fixes for live financial and accounting workflows.
- Implemented ETL-style data processing logic and supported **CI/CD pipelines**, monitoring, and testing for production stability.
- Worked across a distributed backend stack using **Python, node.js and SQL**.

Part-Time (Teaching)

Tangjeong, South Korea

2025(Sep)-2026(Feb)

- Worked as an English teacher in **Premium L ACADEMY** (Elementary and Middle Students)
Baebang, South Korea 2025(Jan-July)
- Worked as an English language Assistant in **더상승영어학원 ACADEMY** (Elementary and Middle and High Students)

- Worked as an English teacher in **SAMSUNG ENGLISH ACADEMY** (Elementary and Middle Students)

Education

Masters in computer science and engineering

[SunMoon](#)

Asan, South Korea

03/2024 - 02/2026

- Fintech • Blockchain • Special Topics in Web Services • Advanced Topics in Information and Data Management
- Research Ethics and Thesis Writing • Special Topics in Reinforcement Learning • Advanced decision-making systems

Bachelor of Science

[UET](#)

Lahore, Pakistan

10/2018 - 11/2022

BS in Computer Science Courses

- MAD-Mobile App Development • W T-Web Technologies • CVIP-Computer Vision and Image processing
- Artificial Intelligence (AI) • Databases System • Machine Learning (ML) • Data Structures and Algorithms • Compiler Construction

Projects

- **Anomaly-detection system:** Built a modular anomaly detection platform with Streamlit UI, implementing multiple ML/DL models, evaluation pipelines, and ensemble methods. ([GitHub](#))
Tech: Python, Scikit-learn, TensorFlow/PyTorch, Pandas, Streamlit
- **Built GPT's core architecture:** self-attention, causal masking, transformer blocks, from PyTorch tensors, not a fine-tuned checkpoint or library call. ([GitHub](#))
- **E-PLANNER:** Web Based 2D and 3D Home Designing — Final Year Project
A website for interior and exterior designing for different buildings using React JS, HTML, JS and CSS and Firebase on backend
- **Internet of Things:** ([GitHub](#))
E-commerce website: Frontend of a wristwatch shop website of different brands using HTML CSS and Java Script for frontend and python for the backend.